

Total No. of Questions : 12]

SEAT No. :

PA-344

[Total No. of Pages : 3

[5927]:228

**B.E. (Mechanical Engineering)
INDUSTRIAL ENGINEERING**

(2015 Pattern) (Semester - II) (Elective - III) (402049B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of programmable calculator is not permitted.
- 5) Assume suitable data, if necessary.

Q1) What is productivity? Discuss in details factors that affect productivity. [8]

OR

Q2) Define Industrial Engineering. Explain in brief various function of Industrial Engineering. [8]

Q3) Write a short note on (any two) . [8]

- a) Multi activity chart
- b) SIMO chart
- c) Stringdiagram

OR

Q4) What is method study? Explain symbols used in recording the activity. [8]

Q5) What is work measurement? Explain different allowances used in standard time calculations? [8]

OR

P.T.O.

Q6) A work study was conducted in a machine shop following data has been recorded. [8]

- a) No of observation = 2000
- b) No of activity idle = 500
- c) Ratio between manual and machine = 3:1
- d) Performance rating = 85%
- e) No of pieces produced = 120
- f) Duration of study = 60 hrs.

Calculate standard time per piece assuming 10% relaxation and 3% contingency allowance.

Q7) a) Explain the functions of PPC. [8]

b) Write note on MRP, MRP-II. [8]

OR

Q8) a) Write a note on forecasting methods. [8]

b) Explain concept of ERP with block diagram. [8]

Q9) a) Explain in detail factors affecting on selection of location for the industrial plant. [7]

b) Write a short note on : [8]

- i) ABC analysis
- ii) Assembly Line Balancing

OR

Q10)a) Estimate the sales forecast for the year 2000, using three month moving average and exponential smoothing forecast. Take $\alpha = 0.5$. [8]

Year	1995	1996	1997	1998	1999
Sale Rs. ($\times 10^5$)	180	168	159	170	188

b) What are different principles of material handling? Describe any four material handling equipment commonly used in industries. [7]

Q11)a) Write short note on : **[8]**

- i) Break even analysis
- ii) Push and pull systems

b) A company has given following information : **[7]**

Data	Rs.
Selling price	20/- unit
Variable cost	5/- unit
Fixed cost	50,000/-

Determine :

- i) BEP
- ii) Sales when it is desirable to earn profit of Rs. 20,000/-

OR

Q12)a) Write a short note on (any two) : **[8]**

- i) KRA
- ii) Cost accounting
- iii) Net present value (NPV)

b) Explain in brief different functions of human resource department.
Comment on Key Result Area and Performance appraisal of employee.

[7]

